Formula 1 Race Action Analysis

DSA 210 Term Project Final Report

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1. Motivation

Some Formula 1 races are full of overtakes and position changes, while others end with everyone basically in the same order they started in. I am a big F1 fan and I wanted to actually look at the data and see what makes a race exciting instead of just relying on a gut feeling.

In this project I define how much action a race has as the average absolute change in position from grid to finish, taken across all classified drivers in that race. Using seven seasons of races from 2018 to 2024, I wanted to answer two main questions. First, which conditions (weather, circuit type, temperature, pit activity) are linked to more action. Second, can a model predict whether a race will be high action using only information that is known before the race starts.

2. Data Source

All the race data was collected with the FastF1 Python library (https://theoehrly.github.io/Fast-F1/), which gives access to the official F1 live timing feed. For every race in the 2018 through 2024 seasons I pulled:

- Session results, including grid position, finishing position, classification status, and laps completed.
- Lap data, which I aggregated per race into pit stop counts and the number of different tyre compounds used.
- Weather data over the whole race: air temperature, track temperature, humidity, wind speed, and a rainfall flag.

I also made a small reference table by hand with the length in km for each circuit, number of turns, and type (street or permanent). I got this info from FIA documents and formula1.com. After joining everything together I ended up with 148 races and 26 columns, saved as data/processed/race_level_data.csv. The raw FastF1 responses are cached locally so the whole pipeline runs fast after the first download.

3. Data Analysis

The analysis lives in two Jupyter notebooks that share helper code in the src/ folder.

3.1 Exploratory Data Analysis

In notebook 01 I first looked at the distribution of the action metric and then broke it down by season, circuit, and circuit type. I made histograms, box plots, scatter plots, and a Pearson correlation heatmap across all the numeric race features. All the figures are saved under reports/figures/.

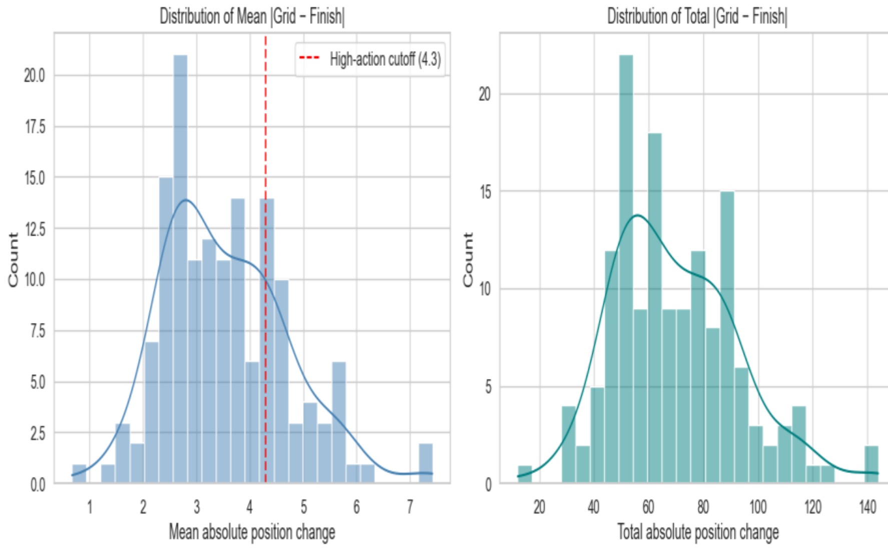

Figure 1. Distribution of the action metric across the 148 races.

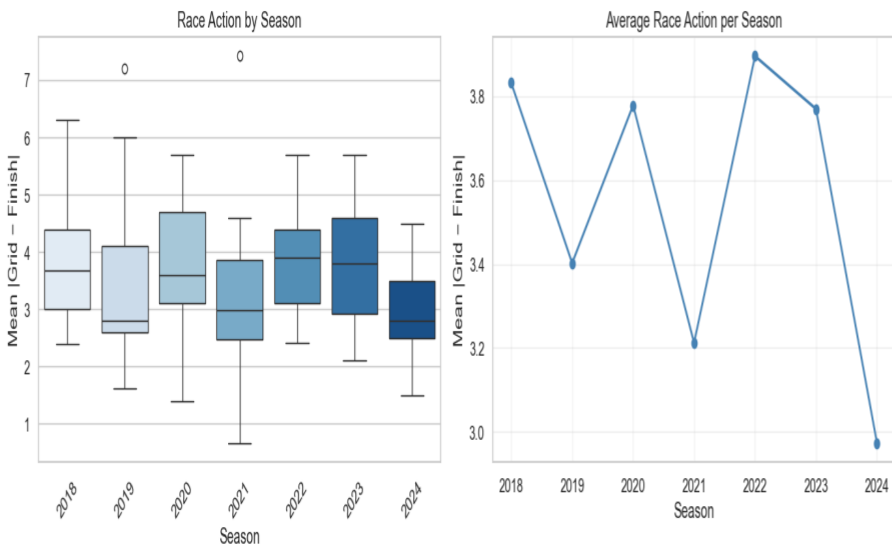

Figure 2. Mean position change broken down by season.

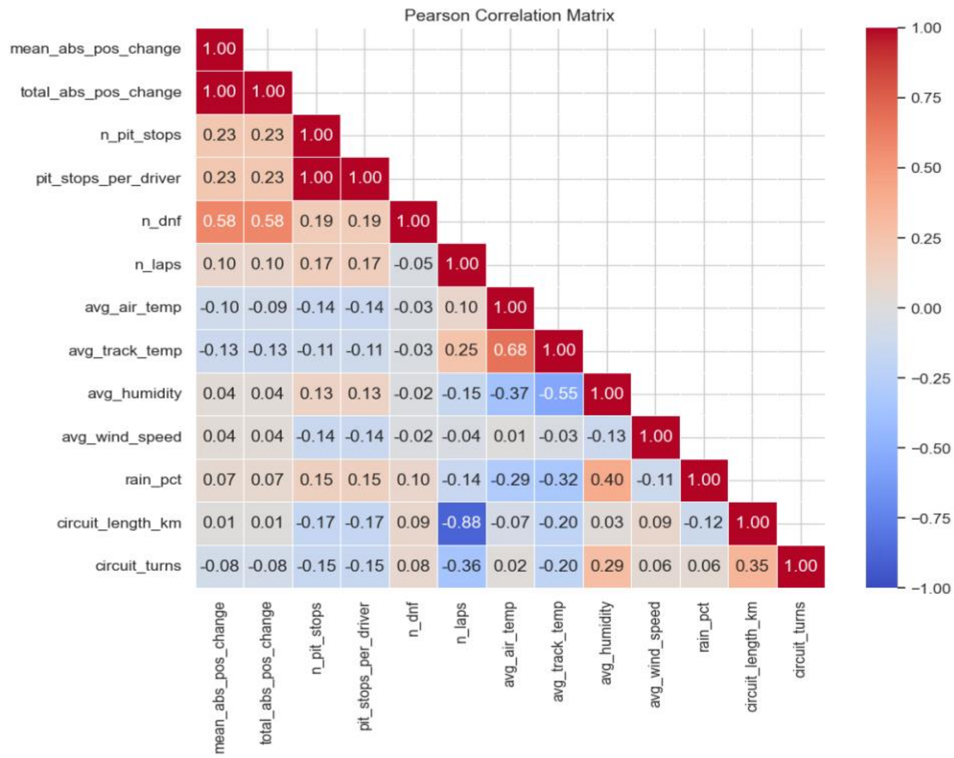

Figure 3. Pearson correlations among the numeric race features.

3.2 Hypothesis Tests

I tested four hypotheses at a significance level of 0.05. For comparing the means of two groups I used independent samples t tests, and for checking whether two numeric variables move together I used the Pearson correlation coefficient.

Hypothesis	Test	p value	Significant at 0.05?
H1: Wet vs dry races	t test	0.2105	No
H2: Street vs permanent circuits	t test	0.9231	No
H3: Air temperature vs action	Pearson r	0.2219	No
H4: Pit stop count vs action	Pearson r	0.0059	Yes (r = +0.225)

3.3 Machine Learning

In notebook 02 I trained two leak free Random Forest models on the same set of features. The first is a classifier that predicts whether a race is in the top 25% for action (40 out of 148 races). The second is a regressor that predicts the continuous action metric directly. The features include season, round, number of starters, classified drivers, DNFs, weather averages, rainfall percentage, circuit length and turns, circuit type, location, and country.

I deliberately left out any feature that is computed from grid to finish positions, because otherwise the model would be cheating by using information derived from the label. Categorical features were one hot encoded and numeric features were filled in with the median when missing. I evaluated both models with 5 fold cross validation and with a separate held out test split.

Model	Cross validated score	Held out test score
Random Forest classifier (high action or not)	ROC AUC = 0.705 (5 fold)	ROC AUC = 0.722
Random Forest regressor (mean abs position change)	R squared = 0.144 (5 fold)	R squared = 0.237, MAE = 0.92 positions

4. Findings

Out of the four hypotheses, only one came out as statistically significant. The number of pit stops in a race is positively correlated with action (r = +0.225, p = 0.0059), so races with more pit stops really do tend to have more position changes.

The other three tests came back as not significant. Wet races had a slightly higher average action than dry ones (3.74 vs 3.47) but the p value was 0.21, which could easily be random noise. Street and permanent circuits were almost identical on the action metric (p = 0.92). Air temperature had basically no linear relationship with action (r = -0.10, p = 0.22). This was a bit surprising, because people usually assume that wet races and street circuits are the chaotic ones, but at least at the seasonal level the data does not really support that intuition.

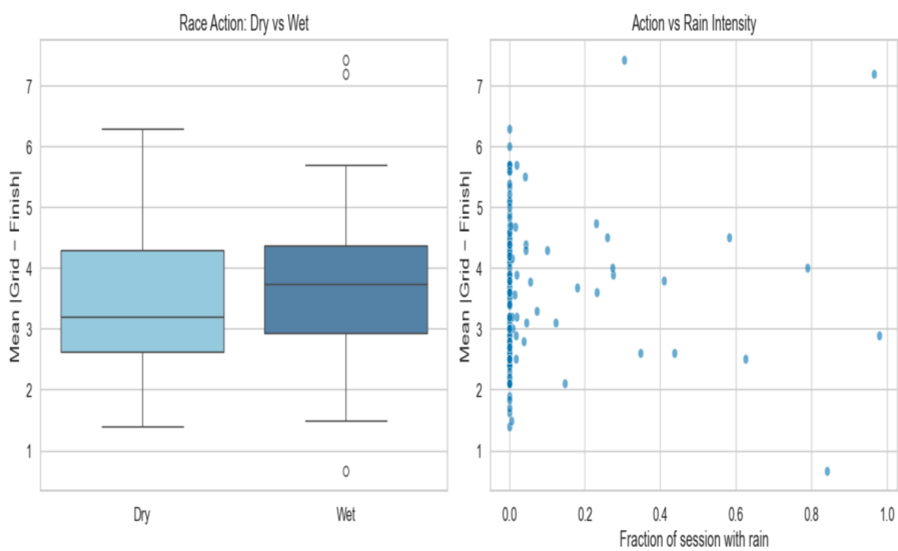

Figure 4. Wet versus dry races. The two distributions overlap heavily.

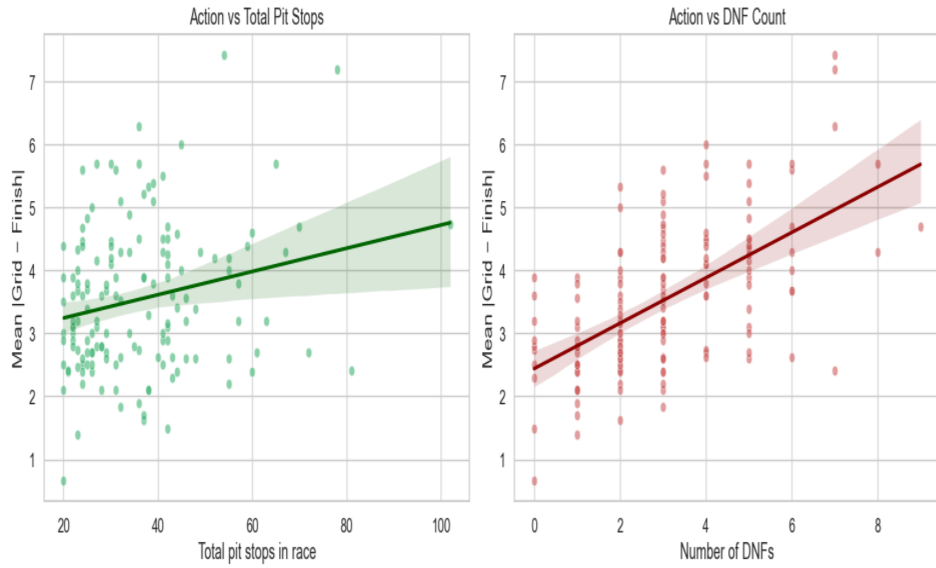

Figure 5. Pit stop count and DNFs both track race action, with DNFs being the stronger signal.

The machine learning models tell a similar story. Using only pre race features and weather, the classifier got a cross validated ROC AUC of 0.705 and a held out AUC of 0.722, which is clearly better than random guessing (0.5). The regressor got an R squared of 0.144 in cross validation and 0.237 on the held out set, with a mean absolute error of about 0.92 positions per driver.

In both models the most important feature was the number of retirements in the race (n_dnf), followed by track and air temperature, humidity, and the physical features of the circuit (number of turns, length). DNFs are partly a result of what happens during the race rather than a real cause, but the pattern still suggests that races with more retirements end up being more eventful. Incidents and mechanical failures shuffle the order in ways that the regular pre race conditions on their own do not.

5. Limitations and Future Work

There are a few things that limit these results.

- Mean absolute position change is a proxy for overtaking. It misses moves that are made and then reversed during the race, and it treats all position changes the same whether they happen on track or through pit strategy.
- Weather data is sampled at fixed points around the circuit, so brief showers that only affect part of the track might not show up in the averages even if they really do change the race.
- Things that clearly shake up races, like safety cars, virtual safety cars, red flags, and race control penalties, are not in the dataset at all. This probably explains a lot of the variance that the model cannot capture.
- 148 races is still a fairly small sample for machine learning. The R squared of 0.14 to 0.24 should be read as a rough signal, not as something you can trust to rank individual races precisely.
- Random Forest feature importance can be biased toward features with many categories or features that are correlated with each other, so the high importance of n_dnf should be read as DNFs being strongly associated with action, not as a clean causal claim.

For future work I would like to add lap by lap overtake data from external trackers so the action metric is less dependent on grid to finish positions. Adding race control event logs (safety car deployments, penalties) as features would probably help a lot too. Including driver and team identity could separate field effects from circuit and weather effects. Finally, extending the dataset to the 2025 season and to sprint races would roughly double the sample size.

6. Reproducibility

All the code, data, and figures are on GitHub at:

https://github.com/alibi03/Formula-1-Race-Action-Analysis-DSA-210-Term-Project-

To reproduce the analysis, run notebooks/01_data_eda_hypothesis.ipynb from top to bottom. This rebuilds the race level dataset and reproduces every figure and hypothesis test in this report. Then run notebooks/02_ml_models.ipynb for the machine learning results. The milestone1 and milestone2 git tags point to the exact commits I used for the two interim submissions.